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Using LSTM with Trajectory Point Correlation and Temporal Pattern Attention for Ship Trajectory Prediction

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Abstract: Accurate ship trajectory prediction is crucial for real-time vessel position tracking and maritime safety management. However, existing methods for ship trajectory prediction encounter significant challenges. They struggle to effectively extract long-term and complex spatial-temporal features hidden within the data. Moreover, they often overlook correlations among multivariate dynamic features such as longitude (LON), latitude (LAT), speed over ground (SOG), and course over ground (COG), which are essential for precise trajectory forecasting. To address these pressing issues and fulfill the need for more accurate and comprehensive ship trajectory prediction, we propose a novel and integrated approach. Firstly, a Trajectory Point Correlation Attention (TPCA) mechanism is devised to establish spatial connections between trajectory points, thereby uncovering the local trends of trajectory point changes. Subsequently, a Temporal Pattern Attention (TPA) mechanism is introduced to handle the associations between multiple variables across different time steps and capture the dynamic feature correlations among trajectory attributes. Finally, a Great Circle Route Loss Function (GCRLoss) is constructed, leveraging the perception of the Earth's curvature to deepen the understanding of spatial relationships and geographic information. Experimental results demonstrate that our proposed method outperforms existing ship trajectory prediction techniques, showing enhanced reliability in multi-step predictions.



Citation: Zhou, Y.; Guo, H.; Lu, J.; Gong, Z.; Yu, D.; Ding, L. Using LSTM with Trajectory Point Correlation and Temporal Pattern Attention for Ship Trajectory Prediction. *Electronics* **2024**, *13*, 4705. <https://doi.org/10.3390/electronics13234705>

Academic Editor: George A. Tsihrintzis

Received: 16 October 2024

Revised: 22 November 2024

Accepted: 26 November 2024

Published: 28 November 2024



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Keywords: ship trajectory prediction; AIS; LSTM; deep learning

1. Introduction

As the volume of cargo transported by sea and the number of vessels in operation have continued to grow, shipping has become a critical component of global trade, representing 90% of total international commerce [1]. Nonetheless, this expansion brings inherent risks, including ship collisions [2], piracy [3], and oil spills [1]. To safeguard ship navigation and optimize waterway usage, accurately determining vessels' locations and movement directions is crucial for effective marine monitoring [4]. Since 2002 [5], many ships have been equipped with the Automatic Identification System (AIS) [6]. AIS dynamically gathers crucial vessel information, including LAT, LON, SOG, heading, and COG [7–9], enabling real-time monitoring of the vessel's operational status and performance [10]. Constructing an appropriate prediction model to learn from historical trajectory data obtained from AIS onboard ships facilitates the accurate prediction of the target trajectory [11]. Accurate trajectory prediction in complex scenarios is essential for maritime safety and plays a critical role in advancing intelligent shipping technologies [12].

Currently, trajectory prediction methods generally combine AIS data with physical models and data-driven approaches [13–15]. These methods can be broadly classified into two categories: traditional methods and advanced methods based on machine learning and deep learning. To better understand the main findings in the field and identify the

gaps in previous research, we have systematically classified and tabulated previous works, along with their key parameters, advantages, and disadvantages, as shown in Table 1.

Table 1. Advantages and disadvantages in existing ship trajectory prediction algorithms.

Previous Works		Pros	Cons
KF		Capable of handling dynamic data in real time and taking uncertainties into account, with acceptable computational efficiency.	Limited by linear assumptions and reliance on accurate system models.
RF		Good at handling high-dimensional data and resistant to overfitting, with relatively good interpretability.	Complex model structure and sensitive to outliers.
SVR		Strong in generalization and overfitting resistance and capable of handling small samples and data with various distributions.	Difficult to select kernel functions and high computational complexity.
RNNs	LSTM	Able to handle long sequences, remember long-term dependencies, and perform automatic learning.	Long training time and prone to overfitting.
	GRU	Simple structure, fast computation, and ability to handle long sequences.	May lose information and have limited generality.
	BiLSTM	Capable of capturing information from both directions of a sequence and handling long sequences.	High computational cost and prone to overfitting.
Transformer		Parallel computing ability, ability to capture long-distance dependencies, and strong generality.	Requires a large amount of resources and ignores sequence order information.
Hybrid methods		Combines the advantages of multiple methods and is adaptable to complex scenarios.	Complex parameter adjustment and difficult to interpret.

Traditional methods integrate physical models and data-driven models for prediction, encompassing the Kalman filter (KF) [16,17], Gaussian mixture model (GMM) [18], support vector machine (SVM) [19–21], and other technologies to enhance prediction accuracy. Due to the restricted capacity of these methods in handling uncertain factors, nonlinear motion, and non-static characteristics, traditional methods relying solely on motion equations are unable to achieve precise predictions.

Subsequent studies have introduced more sophisticated methods based on the variances in data complexity and model generalization ability. These methods can be categorized into traditional machine learning models and deep learning models. They utilize historical data to extract and track patterns and features, strengthening adaptability to diverse ship movement patterns and environmental changes.

Supervised learning depends on labeled data, with random forests (RF) [22] and support vector regression (SVR) being commonly used to make effective predictions, while unsupervised learning, including methods such as clustering and dimensionality reduction, can identify inherent patterns in historical trajectory data.

The rapid advancement of deep learning algorithms has led to significant progress in ship trajectory prediction techniques. Recurrent neural networks (RNNs) [23–27], bidirectional recurrent neural networks (BiRNNs) [28–31], convolutional neural networks (CNNs), attention mechanisms (AMs), and Transformers [32–38] have been applied. However, deep learning methods also have limitations, such as difficulties in addressing long-term dependencies and effectively utilizing prior knowledge.

However, due to the limitations of deep learning in addressing long-term dependencies and the utilization of prior knowledge, some scholars have proposed a hybrid method combining deep learning with other methods to solve trajectory prediction problems, which can be divided into two mainstream approaches. The first method involves clustering historical trajectory information to obtain several similar clusters and training the model for each one [39]. The second approach involves extracting key waterway features and using them to refine the prediction results generated by neural networks [14]. All the above methods first extract trajectory information for the prediction, but this knowledge is not fully integrated into the neural network, making it difficult to obtain potential geographic information.

Based on a comprehensive review of previous works, it is evident that there are still gaps in existing ship trajectory prediction methods. Existing trajectory prediction models have limitations in capturing long-term dependencies and accurately representing the dynamic correlations between multiple features, such as longitude, latitude, speed, and course. These models often fail to address the complexities introduced by spatial-temporal dynamics, leading to less reliable predictions, especially in multi-step forecasts. In this work, we propose a novel approach that integrates TPCA, TPA, and GCRLoss to address these challenges. Our method not only improves the accuracy of trajectory prediction but also provides a more robust solution for real-time maritime safety monitoring and navigation. The key contributions are summarized as follows:

Trajectory Point Correlation Attention (TPCA): We design a TPCA mechanism to capture the variations between trajectory points during ship motion. By establishing spatial correlations among trajectory points, nuanced local trends within trajectories are effectively revealed, thereby bolstering the model's ability to discern and predict local trajectory dynamics.

Temporal Pattern Attention (TPA): We introduce a TPA mechanism combined with an LSTM model to comprehensively analyze the correlations among multiple variables in ship trajectory data across various time steps. This approach takes into account the spatial positions of trajectory data and effectively captures dynamic feature correlations over time.

Curvature-aware Great Circle Route Loss (GCRLoss): We construct a GCRLoss function based on the perception of the Earth's curvature. By integrating geographical knowledge into the modeling, we can calculate errors in geo-locations more accurately.

The rest of this article is organized as follows. Section 2 provides an overview of the work and the limitations of current AIS-based ship trajectory prediction. Section 3 provides a detailed introduction to the improved algorithm. Section 4 describes the dataset and experimental settings. The performance of this method is also evaluated. Finally, this paper also discusses the advantages and disadvantages of the model and future research directions in Section 5, where the study is summarized.

2. Related Works

In this section, we will briefly review trajectory prediction methods that utilize deep learning and hybrid methods, as they are closely related to our work.

2.1. Ship Trajectory Prediction Using Deep Learning-Based Methods

Deep learning models have unique advantages in using large-scale data of a similar type for prediction. These methods primarily use neural networks to extract contextual information from trajectory data, analyze related features, and predict future ship trajectories [40]. In the field of ship trajectory prediction, a series of deep learning techniques have been used with historical AIS data, including LSTM, Gated Recurrent Unit (GRU) networks, BiLSTM, and Transformers.

2.1.1. Ship Trajectory Prediction Method Based on the RNN Model

RNN is a general term encompassing RNN, LSTM, GRU, and other recurrent neural network models. RNNs are specifically designed to handle sequential data and have found

extensive applications in various fields, including ship trajectory prediction. RNNs, in particular, excel at handling extensive discrete time-series data and are uniquely suited to capturing temporal dependencies. With this feature, the RNN can realize the correlation features of trajectory data and fully exploit the internally embedded information.

Additionally, replacing the hidden layer of the RNN with an LSTM network unit has yielded better results for practical trajectory prediction models. Numerous researchers have conducted studies focusing on LSTM and its variants. Tang et al. [41] devised an LSTM-based model to predict ship trajectories within Tianjin Port, China. Their model comprises two LSTM layers and utilizes ship observation data collected over 10-minute intervals as input. Mehri et al. [42] proposed a hybrid method that combines a curve model with a context-aware LSTM (CLSTM) network, incorporating the influence of the geographic environment to ensure logical consistency in a compressed trajectory. Liu [43] introduced an interactive vessel trajectory prediction framework based on LSTM named QSD-LSTM. This framework incorporates a quaternion ship domain (QSD) to prevent unnecessary collisions between adjacent vessels. Tian [44] addressed the challenges of inaccurate and time-consuming vessel trajectory predictions by proposing a D-LSTM network. This network captures differences and trends in sequential data more precisely through its unique structure and algorithms, enabling better prediction in complex maneuvers. Building on the D-LSTM framework, Li [45] introduced a trajectory correction function. It analyzes deviations between target and sample features and adjusts them, resulting in Ref-DLSTM, which integrates them well, improving accuracy. Ma [46] discovered that vessel motion is often highly correlated with past movements and proposed an ALSTM model. Traditional LSTM has issues like poor long-term dependency capture. ALSTM overcomes this by using skip connections and adaptive memory modules. In tests, ALSTM showed higher accuracy than traditional LSTM. Syed, M.A.B [47] proposed a 1D CNN-LSTM architecture for track association that leverages the strengths of two distinct neural networks to model the spatial and temporal aspects of AIS datasets. Yan [48] introduced QCNN-LSTM. It uses two types of predictive input. Through attention mechanisms, it calculates importance scores and focuses on important parts, reducing information decay during current input processing. Wang [49] proposed a CNN-LSTM-SE model. It uses specific operations to quantify correlations between variables and temporal features, improving prediction accuracy. Li [50] introduced a hybrid approach that combines Graph Attention Networks (GAT) with LSTM networks. The GAT-LSTM model significantly enhances trajectory prediction accuracy and robustness by incorporating spatiotemporal features during the prediction process. Li [51] put forward the ACoAtt-LSTM method. In this approach, data encoding is employed to format the data in an organized manner. Meanwhile, the attention modules play a crucial role by concentrating on different attributes and assigning corresponding weights to them. Moreover, the LSTM network is utilized to handle the temporal information effectively. The coordinated operation of these components brings about improvements in various aspects, thereby effectively addressing the complex issues related to vessel motion. Guo [45] introduced the VI-LSTM model. It uses algorithms to map and combine environmental factors, vessel motion, static characteristics, and influencing factors. This integration improves performance by capturing real-world complexity. Wang [52] proposed a PSO-LSTM model. PSO initializes particles representing combinations of hidden units and learning rates. They move and update to converge to the optimum. This optimization improves model performance, e.g., higher accuracy in tests.

In time-series forecasting tasks, unidirectional neural network structures only propagate states from front to back. However, for a more accurate prediction, it would be more beneficial to link the output at the current time to the state at the previous and next time steps. Bidirectional neural networks provide a solution for this task by linking the current output to the preceding and succeeding time-step states. Bi-LSTM consists of two independent LSTMs capable of capturing past and future information. Li [28] addressed the issue of local optima in machine learning algorithms by proposing an improved Bi-LSTM model. This model incorporates Rectified Adaptive Moment Estimation (RADam) and Lookahead

techniques, where RAdam helps in optimizing the learning process to avoid getting stuck in local optima, and Lookahead provides a more forward-looking perspective to further refine the model's performance. Ma [29] combined Bi-LSTM with attention mechanisms. By leveraging the attention mechanisms to assign different weights to various spatiotemporal elements of the data, it effectively captures the spatiotemporal dependencies of behaviors and accurately assesses their impact on future risks. Zhang [30] proposed a novel method for ship trajectory prediction, integrating an LSTM autoencoder, an attention mechanism, and a bidirectional LSTM (AABiL) structure. Kristian Aalling Sørensen [5] introduced the Bidirectional Long Short-Term Memory Mixture Density Network (BLSTM-MDN), a deep learning model designed to accurately capture the underlying distribution of ship trajectories. To address the challenges of hyperparameter design, Jia [31] optimized the hyperparameters of the Attention-BiLSTM network using the Whale Optimization Algorithm (WOA).

The GRU, another variant of LSTM, was also proposed to overcome the difficulties of long-term memory and the gradient in backpropagation. Cen [53] proposed a CNN-GRU model with a hybrid attention mechanism (AM). Li [54] used the Wavelet Transform-Gated (WTG) model, integrating wavelet transforms, Temporal Convolutional Networks (TCNs), and GRUs to enhance ship path prediction. Ma [55] introduced the dynamic spatiotemporal model TG-PGAT (Temporal Convolutional Network–Bidirectional GRU–Pearson Correlation Coefficient Graph Attention Network) for predicting traffic flow in port waterways. Liu [56] proposed the CNN-RGRU-Attention model, integrating a CNN, Attention Mechanisms, and GRUs. To overcome the limitations of traditional neural networks in extracting features from ship trajectory sequences and achieving the required precision for deep-channel navigation, Dong [57] proposed the enhanced model CNN-MTABiGRU with a time attention mechanism and the Mish activation function. Lin [58] developed the hybrid model TTCN-Attention-GRU (TTAG) for trajectory prediction by integrating the TTCN framework with attention mechanisms and GRU networks. Chen [59] proposed a trajectory prediction method using BiGRU and trajectory direction vectors (TDVs) enhanced by an attention mechanism.

2.1.2. A Ship Trajectory Prediction Method Based on the Transformer Model

The Transformer model is capable of predicting future ship positions and movement directions by learning patterns and trends from historical trajectory data. It effectively captures the dependency relationships and contextual information across different time steps, predicting future ship trajectories more accurately. Zhao [32] proposed a Transformer model to handle chaotic time-series prediction tasks, successfully integrating the attention mechanism of deep learning into the analysis of chaotic time series. This approach addresses the intrinsic challenges of nonlinearity and long-range dependencies within chaotic sequences. Xue [33] introduced G-Trans, a hierarchical Transformer architecture that utilizes GRU transformations and feature clustering for ship trajectory prediction. Additionally, a multi-stage deep learning context-aware method known as DCA-MSDL was proposed [34], which builds a deep learning architecture based on the inverted Transformer (iTransformer). NGUYEN [35] developed TrAISformer, an enhanced Transformer network designed to extract long-term temporal patterns from AIS ship trajectories within a rich spatial framework, allowing for the prediction of ship positions several hours ahead. Jiang [36] investigated a novel trajectory prediction method by combining LSTM modules with Transformer models. Huang [37] presented a deep learning approach for ship trajectory prediction that incorporates discrete meteorological data via the TripleConvTransformer framework. Lastly, Qiang [38] proposed MSTFormer, a motion-inspired ship trajectory prediction method founded on the Transformer architecture.

2.2. Ship Trajectory Prediction Using Hybrid Methods

Deep learning-based methods mainly rely on features learned from AIS data, making it challenging to utilize prior domain knowledge or expert experience directly. However, this

knowledge plays a crucial role in enhancing prediction accuracy in many cases. Therefore, some studies have attempted to integrate neural networks with other methodologies to leverage domain knowledge for predictive assistance. Some methods use clustering first to classify the area and then train neural networks for prediction in the corresponding areas [38]. Murray et al. [39] introduced clustering with deep learning, consisting of three main components. The initial part focuses on clustering, where an encoder–decoder is employed to extract features from historical ship track data. These features are then organized into clusters using a hierarchical DBSCAN method. The obtained representation is subsequently input into the hierarchical DBSCAN clustering algorithm. In the next stage, the Bidirectional GRU (BiGRU) model [60] is utilized to categorize new trajectories into several potential clusters. The clusters identified in this stage are then employed to predict trajectory groupings. Xi et al. [61] introduced a prediction model for similarity search based on LSTM. The Dynamic Time Warping (DTW) technique is initially employed to identify the trajectory that closely aligns with the reference [62]. Subsequently, LSTM is constructed using historical AIS data to predict spatial distance and capture the vessel motion trend. Suo [63] introduced a hybrid approach that combines unsupervised DBSCAN clustering with a GRU model. Before training the GRU model, AIS trajectory data underwent preprocessing using the DBSCAN algorithm to identify the main ship trajectory areas and remove redundant data via similarity analysis. Furthermore, some approaches utilize extracted waterways or analogous historical trajectories to enhance neural network predictions. Alizadeh [64] introduced three trajectory models centered on similarity search, each emphasizing the distance between the target and sample trajectories to improve prediction accuracy by minimizing this distance. While these methods utilize knowledge to assist network prediction, they fail to fully integrate this knowledge into the neural network and cannot deeply learn dynamic knowledge.

In conclusion, this paper presents a comprehensive exploration of ship trajectory prediction methods. In the Related Works Section, we surveyed the diverse landscape of existing approaches, including deep learning-based methods such as RNNs and their variants (LSTM, Bi-LSTM, and GRU), as well as the Transformer model, along with hybrid methods that attempt to integrate neural networks with other techniques to leverage domain knowledge. These prior efforts have laid a solid foundation and provided valuable insights for our research. Building on this foundation, our proposed methodology addresses the ship trajectory prediction problem with a novel approach. We define the problem by representing ship trajectories using AIS data and formulating the prediction task as mapping historical trajectory data to future predicted states. The model architecture we present consists of three key components: Multi-head TPCA, a TPALSTM Network, and a GCRLoss function.

3. Methodology

3.1. Problem Definition

In this paper, we utilize the Maritime Mobile Service Identity (MMSI) from AIS data as the ship's unique identifier. Ship trajectories for a series of time periods are represented by T_n [51], as shown in Equation (1):

$$T_n = (C_{t_1 t_2}, \dots, C_{t_{n-1} t_n})_i = (P_{t_1}, P_{t_2}, \dots, P_{t_n})_i. \quad (1)$$

In this context, i represents the MMSI number, $C_{t_i t_j}$ represents the segmented trajectory, and P_{t_n} denotes the four feature vectors within the AIS data of the ship at time t_n , as shown in Equation (2):

$$P_{t_n} = [lon, lat, cog, sog]_{t_n}. \quad (2)$$

where lon represents longitude, lat represents latitude, cog represents course over ground, and sog represents speed over ground.

$$Y_{t_{n+1}} = F(P_{t_1:t_n}). \quad (3)$$

Figure 1 illustrates the input of historical AIS data $P_{t_1:t_n}$ of the ship from time t_1 to t_n into the prediction model function F . This process outputs the predicted ship information $Y_{t_{n+1}}$, as outlined in Equation (3). $Y_{t_{n+1}}$ signifies the predicted ship state information at moment t_{n+1} . The predicted data $Y_{t_{n+1}}$ are then compared with the actual historical data $P_{t_{n+1}}$.

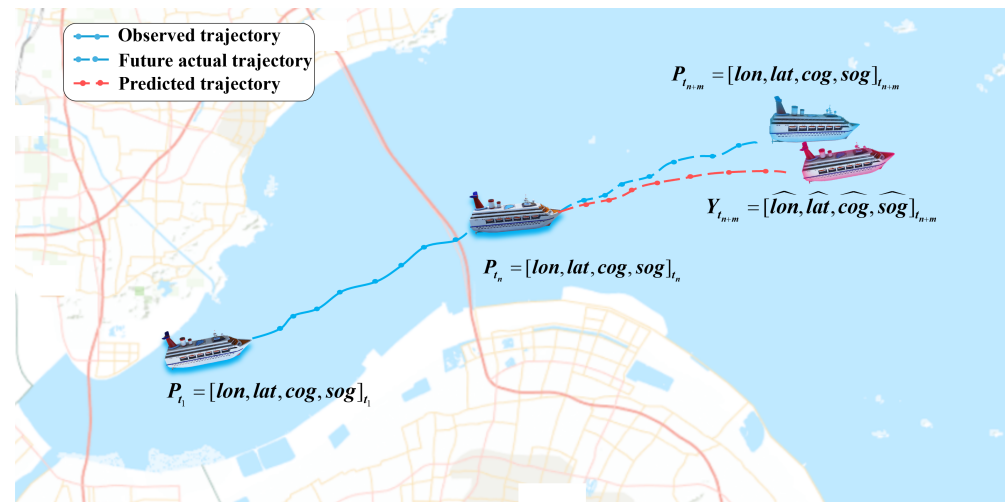


Figure 1. The historical trajectory of a ship at sea in the first n moments predicts the trajectory at $n + m$ moments.

3.2. Model Architecture

Expanding on the previously mentioned trajectory prediction methods, this paper puts forward an approach for predicting trajectories, as depicted in Figure 2. Our method is structured with three principal components: Multi-head TPCA, a TPALSTM Network, and a GCRLoss function that takes into account the Earth's curvature. The TPCA mechanism is designed to decipher the relationships among trajectory points and spot local trends within the trajectory. Meanwhile, TPA is incorporated to strengthen the association between multiple variables of the trajectory data across multiple time steps. Finally, the model is optimized using the great circle distance loss function to assess the distance between predicted and actual trajectory positions. To facilitate a deeper comprehension of these components and their functions in our method, we first conducted a comprehensive analysis, as shown in Table 2, and then introduced each component separately. The table not only highlights the similarities and differences between each component and previous methods but also showcases their respective main advantages and potential limitations.

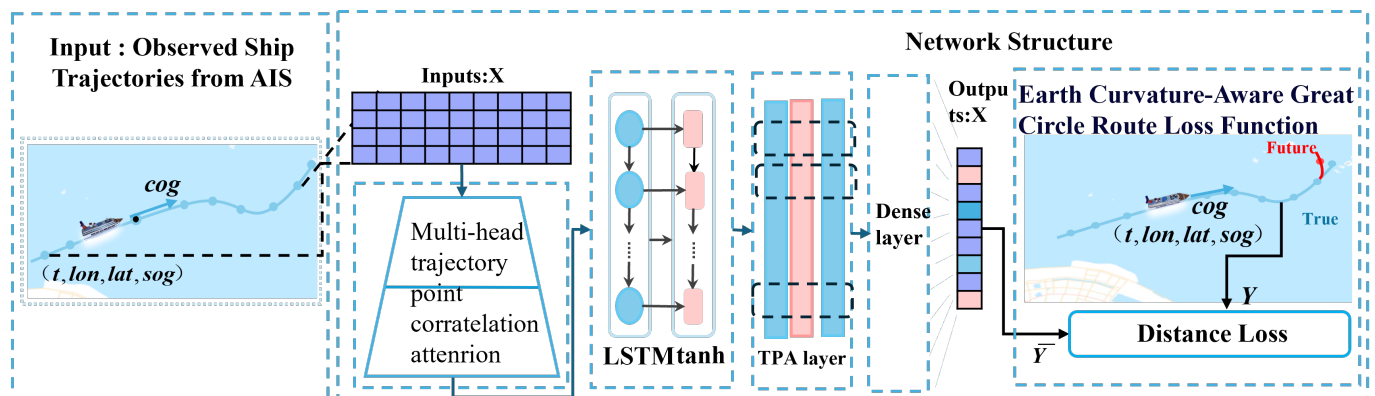


Figure 2. Trajectory prediction model framework.

Table 2. Comparative analysis of components in our method.

Components	Similarities	Main Advantages	Possible Limitations
TPCA	It aims to enhance the understanding of ship trajectories by processing data relationships and exploiting the attention mechanism to extract crucial information from the data.	It not only focuses on the order of trajectory points but also gains a deeper comprehension of the correlations among them. It can identify hidden patterns within complex data structures.	The calculation process is relatively complex, which may increase the computational cost and time when dealing with large-scale data.
LSTM	Similar to previous studies, this study utilizes LSTM to handle time-series data like ship trajectories for prediction.	It can effectively model the long-term dependencies in time-series data and plays an important role in predicting the position of the ship at different moments.	Its ability to handle complex nonlinear relationships is relatively limited, and it may not be able to fully capture some complex variation patterns in ship trajectory data.
TPA	It is also designed to better utilize the information in time-series data and improve the prediction effect through a certain form of attention mechanism.	It can attend to multivariate data such as LON, LAT, SOG, and COG multiple times, enabling a more comprehensive consideration of the dynamic feature correlations among trajectory attributes.	Its adaptability to diverse data scenarios, such as different ship types and different sea areas, may require further verification and optimization.
GCRLoss	With the aim of optimizing the ship trajectory prediction model, an appropriate loss function is designed to adjust the model parameters to improve the prediction accuracy.	It can accurately consider the curvature of the Earth, addressing the fact that the Euclidean distance cannot accurately reflect the distance between the true and predicted positions.	The calculation is relatively complex, and compared with the traditional loss function based on the Euclidean distance, the computational cost may be higher.

3.2.1. TPCA: Trajectory Point Correlation Attention

The attention mechanism in artificial intelligence draws inspiration from human cognitive abilities to selectively focus on specific moments and areas amidst vast datasets. It highlights critical details while filtering out irrelevant information, akin to how humans prioritize and process essential aspects of information. We devise a TPCA to enhance feature extraction from trajectory data.

TPCA, as depicted in Figure 3, enables the model to process a series of ship trajectory points, not only focusing on their order but also gaining a deeper comprehension of their correlations. This is crucial for identifying patterns that might not be intuitive or are concealed within complex data structures.

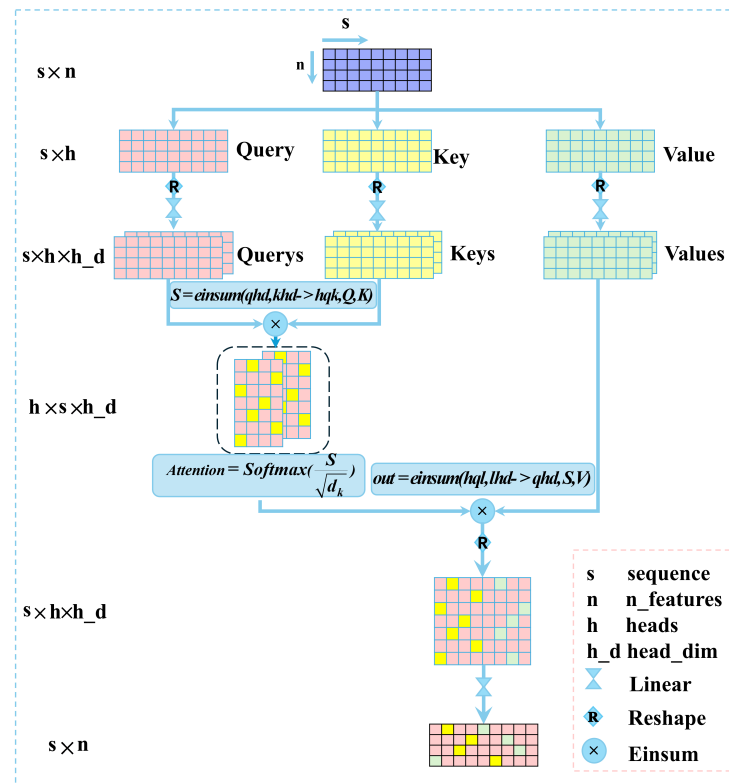


Figure 3. Trajectory Point Correlation Attention.

Each historical trajectory point of the ship is embedded into a high-dimensional space to generate the corresponding query, key, and value. In Figure 3, the query, key, and value are first rearranged to obtain K and $V \in \mathbb{R}^{sequence \times heads \times head_dim}$, and the input X of the attention block is linearly projected $Q = XW_Q$ using $W_Q, W_V, W_K \in \mathbb{R}^{d_k \times d_k}$. Similarly, we can obtain $K = XW_K, V = XW_V$. The attention score S is generated by calculating the similarity between the query and the key, thereby capturing the correlation between different trajectory points. This is accomplished by using the einsum function, which conducts batch matrix multiplication to compute these scores S . The expression for S is defined in Equation (4):

$$S = \text{einsum}(qhd, khd \rightarrow hqk, Q, K), \quad (4)$$

The resulting attention scores (*Attention*) are standardized by the *Softmax* function so that the sum of the attention scores for each element is as shown in Equation (5). This means that the model can decide how much attention to assign to each element in the sequence (which can be understood as a weight assignment). The attention formula is

$$\text{Attention} = \text{Softmax}\left(\frac{S}{\sqrt{d_k}}\right), \quad (5)$$

The standardized attention scores (*out*) are used as weights for the corresponding values to generate a weighted representation of each historical trajectory point, which is a collection of information on all elements as shown in Equation (6).

$$\text{out} = \text{einsum}(hql, lhd \rightarrow qhd, \text{Attention}, V), \quad (6)$$

Finally, the result is passed through a fully connected layer to ensure that the output remains consistent with the input as shown in Equation (7).

$$\text{out} = \text{linear}(\text{heads} * \text{head_dim}, n_features). \quad (7)$$

By designing a trajectory point association attention mechanism in this study, we can enhance the accuracy of predicting the ship's future position. Additionally, this approach enables us to comprehensively explain the prediction results, including identifying which historical trajectory points significantly influence the current prediction.

3.2.2. LSTM Network

According to definition in Section 3.1, vessel trajectories can be considered time-series data. It is well known that an RNN can effectively predict time-series data, contributing to the spatiotemporal trajectory prediction. As a consequence, LSTM, a special kind of RNN, and its extended versions have recently attracted increasing attention in vessel trajectory prediction. Benefiting from the memory and forgetting strategies, LSTM can model long-term dependencies in a sequence of timestamped points. The working mechanism of a typical LSTM framework is displayed in Figure 4. For the sake of simplicity, we continue to use Equation (2) to denote the timestamp at time points t_1 through t_n . The LSTM network then computes the hidden state h_t and the predicted point P_{t_n} at time t_n by recursively calculating the following equations:

$$h_{t_n} = \text{LSTM}(P_{t_n}, h_{t_{n-1}}; \mathbf{W}), \quad (8)$$

$$P_{t_{n+1}} = \mathbf{W}_P h_{t_n} + b_P. \quad (9)$$

Here, $\mathbf{W}$ represents the general definition of the weight matrix set, $\mathbf{W}_P$ denotes the weight matrix between the hidden and output layers, and b_P denotes the bias vector for the output (i.e., predicted point) $P_{t_{n+1}}$. Within each LSTM memory block, the hidden state h can be estimated in detail using the input gate vector i , forget gate vector f , cell state vector c , and output gate vector o . As shown in Figure 3, based on the current input P_{t_n} and output $h_{t_{n-1}}$ of LSTM, the current block input component C_{t_n} is depicted as follows:

$$C_{t_n} = \tanh(\mathbf{W}_c[h_{t_{n-1}}, P_{t_n}] + b_c), \quad (10)$$

where b_c is the bias vector, and $\mathbf{W}_c$ denotes the weight matrices associated with P_{t_n} and $h_{t_{n-1}}$. We then update the vectors i , f , C , and O according to the following formulas:

$$i_{t_n} = \sigma(\mathbf{W}_i[h_{t_{n-1}}, P_{t_n}] + b_i), \quad (11)$$

$$f_{t_n} = \sigma(\mathbf{W}_f[h_{t_{n-1}}, P_{t_n}] + b_f), \quad (12)$$

$$C_{t_n} = f_{t_n} C_{t_{n-1}} + i_{t_n} C_{t_n}, \quad (13)$$

$$O_{t_n} = \sigma(\mathbf{W}_o[h_{t_{n-1}}, P_{t_n}] + b_o). \quad (14)$$

where $\odot$ represents the point-wise multiplication operator. Analogous to the definitions of variables in Equations (11), (12) and (14), $\mathbf{W}$ denotes the weight matrix, $\sigma(\cdot)$ stands for the sigmoid activation function, and each b is the bias vector for the corresponding layer. Finally, the block output h_{t_n} , which combines the current cell vector C_{t_n} and the output gate vector O_{t_n} , can be calculated as follows:

$$h_{t_n} = O_{t_n} \odot \tanh(C_{t_n}). \quad (15)$$

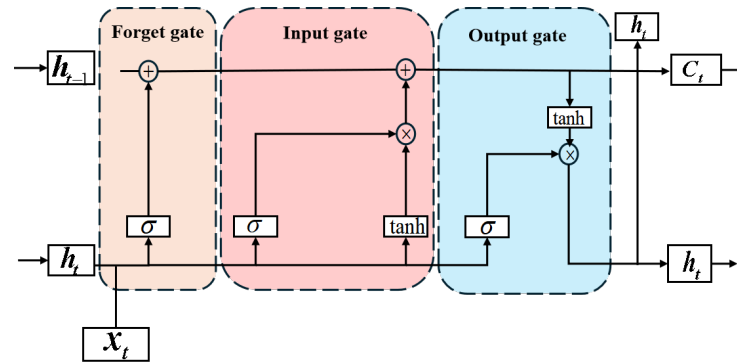


Figure 4. LSTM network.

3.2.3. TPA: Temporal Pattern Attention

Ship trajectory prediction is a type of multivariate time-series prediction involving multiple variables, like LON, LAT, SOG, and COG, at each time step. These variables exhibit complex nonlinear relationships across various time steps, as illustrated in Figure 5. LAT and LON changes determine the ship position's movement, and COG and SOG directly affect their variation trends. For example, increased SOG generally proportionally raises the rates of change in LAT and LON (assuming constant COG). But, SOG changes interact nonlinearly with LAT and LON changes. COG changes also complexly impact the ship's position at different LATs and LONs. This complicates the final prediction results. Moreover, each variable sequence displays unique characteristics and cycles. This makes it challenging to discriminate the focal point. Traditional attention mechanisms focus mainly on information at the current time step and allocate weights evenly across multiple time steps, which limits their ability to capture and predict valuable temporal patterns.

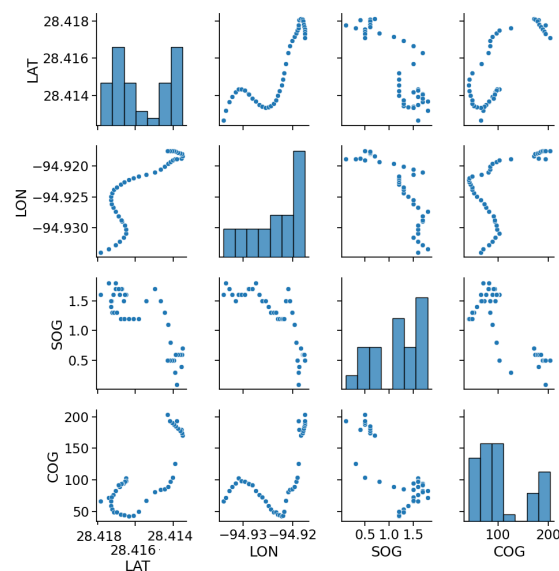


Figure 5. Relationship chart of different ship attributes.

To fully leverage the valuable information in time-series data, this paper introduces TPA [65] into the LSTM prediction model, building upon the TPCA block discussed in the previous section. TPA enables the model to attend to LON, LAT, SOG, and COG data multiple times, facilitating a more comprehensive consideration of the dynamic feature correlations among trajectory attributes. Figure 6 presents a heat map illustrating the relationships among various attribute values. This section elucidates the significance of TPA in ship trajectory prediction.

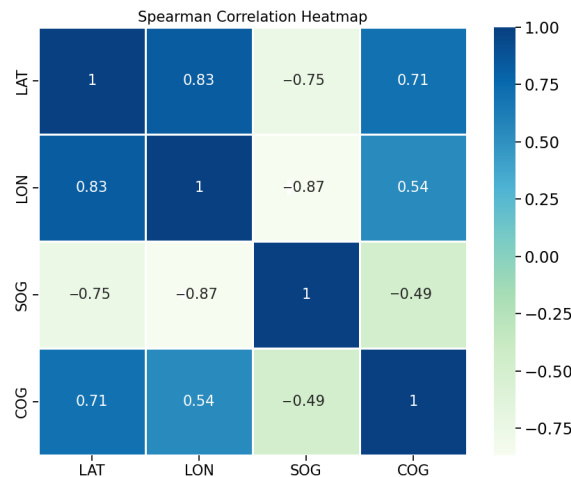


Figure 6. Visualization of the relationships between different ship attributes.

The concept of temporal patterns is introduced as an extension of the traditional attention mechanism to enhance the exploitation of discriminative patterns and features in time-series data. TPA utilizes CNN filters to extract fixed-length temporal patterns from the input data, the scoring function is used to determine the weight of each timing pattern, and finally, the output information is generated according to the weight.

Given the previous LSTM corresponding to the different time inputs of the implicit state matrix $H = (h_{t-w}, h_{t-w+1}, \dots, h_{t-1})$, where w is the time-series length, and h is obtained from the LSTM structure in the previous subsection, each hidden column vector denotes the internal neuron parameters of LSTM at a given time step, while each row vector represents the state of a ship attribute across all time steps. The boxes in the hidden state matrix H represent various one-dimensional convolution kernels. By applying one-dimensional convolution across the m feature convolutions, the time pattern matrix H^C of the variable signal model is extracted as shown in Equation (16):

$$H_{i,j}^C = \sum_{l=1}^w H_{i,(t-w-1+l)} * C_{j,T-w+l}, \quad (16)$$

In this context, C_j represents the j -th filter of length T , where T denotes the maximum length considered, typically denoted by ω . The symbol $*$ denotes the convolution operation. The one-dimensional filter is composed of k convolution kernels, each of which operates along the row vectors of the hidden state matrix. The temporal pattern matrix contains the intricate internal connections and temporal relationships among different sequences, serving as a high-dimensional representation of the complex nonlinear relationships among them. A score function f is used to calculate correlations among ship data attributes as shown in Equation (17):

$$f(H_i^C, h_t) = (H_i^C)^\top W_a h_t, \quad (17)$$

where H_i^C is the i -th row of H^C , and $W_a \in \mathbb{R}^{k \times m}$. The attention weight α_i of each attribute as shown in Equation (18) is

$$\alpha_i = \text{sigmoid}(f(H_i^C, h_t)), \quad (18)$$

Using the obtained attention weights α_i and H^C -weighted summation, one can obtain the context vector v_t as shown in Equation (19):

$$v_t = \sum_{i=1}^n \alpha_i H_i^C, \quad (19)$$

In the formula, n represents the number of features of the input variable x . v_t and h_t are linearly mapped and then summed to produce the final prediction value $y_{t_1+\Delta}$ as shown in Equation (20):

$$y_{t_1+\Delta} = W_{h'}h'_t = W_{h'}(W_h h_t + W_v v_t). \quad (20)$$

where $y_{t-1+\Delta}$ represents the final prediction value; h' is the intermediate variable used to generate the final value; Δ represents the prediction time scale of different prediction tasks; and $W_{h'}$, W_h , and W_v are different weight matrices for the corresponding variables.

The first two sections focus on the mechanisms of attention, specifically TPA and TPCA. A Transformer is also a type of attention mechanism: what are the similarities and differences between them? A commonality among these approaches is that their objective is to process data relationships in order to enhance the understanding of ship trajectories while improving processing capabilities through the application of attention. In terms of distinctions, the TPCA mechanism primarily aims to uncover hidden correlations between track points. In contrast, the TPA mechanism conceals the state matrix surrounding LSTM and employs convolutional techniques to extract temporal patterns. The Transformer architecture emphasizes global dependencies in sequential data processing; it has the capability to simultaneously capture multi-level relationships. Its multi-focus design enhances both data representation and operations by providing greater flexibility and comprehensively capturing features from various perspectives—each approach being suitable for different scenario requirements. In this paper, we propose that a combination of TPA and TPCA trajectory prediction models can more effectively extract correlations between trajectory points, as well as dynamic feature correlations among trajectory attributes.

3.2.4. GCRLoss: Curvature-Aware Great Circle Route Loss Function

The GCRLoss function introduced in this paper computes the loss based on the average great circle distance between the true and predicted ship positions, as shown in Equation (21). Typically, scholars use the mean squared error of Euclidean distance for loss functions in ship trajectory prediction tasks. Subsequently, optimization algorithms like gradient descent are employed to adjust model parameters, aiming to minimize the loss function and optimize the trajectory prediction model. However, using the Euclidean distance has limitations, particularly in geospatial tasks like ship trajectory prediction. Since the Euclidean distance does not account for the curvature of the Earth's surface, it does not accurately reflect the distance between the true and predicted positions. The great circle distance takes into account the curvature of the Earth's surface. In ship trajectory prediction, the model aims to predict a continuous sequence of position coordinates. Using the great circle distance as a loss function accounts for position offsets and uncertainties, thereby enhancing prediction accuracy and reliability. Due to the Earth's irregular spherical shape, the function $R(LAT)$ is used in the formula to calculate the Earth's radius at different locations as shown in Equation (21):

$$R(LAT) = \sqrt{\frac{(A^2 \cos LAT)^2 + (B^2 \sin LAT)^2}{(A \cos LAT)^2 + (B \sin LAT)^2}}, \quad (21)$$

A is the equatorial radius of WGS-84, and B is its polar radius, where $A = 6,378,137.0$ and $B = 6,356,752.3142$. The Havers formula calculates the great circle distance between the real point and the predicted point. The detailed steps for calculating this distance are summarized in Algorithm 1. The corresponding loss function is calculated as follows:

$$L_{GCR} = \frac{\sum_{n=1}^N dis}{N}. \quad (22)$$

Algorithm 1 GCRLoss function.

```

function GCRLOSS(origin, destination, predicted)
  Input:
    origin: (lat1, lon1)                                ▷ Use abbrev. like ‘Origin coords’ if needed
    destination: (lat2, lon2)                                ▷ Dest. coords
    predicted: (predlat, predlon)                            ▷ Pred. coords
  Convert coordinates to radians
    lat1 ← origin.lat ×  $\frac{\pi}{180}$ 
    lon1 ← origin.lon ×  $\frac{\pi}{180}$ 
    lat2 ← destination.lat ×  $\frac{\pi}{180}$ 
    lon2 ← destination.lon ×  $\frac{\pi}{180}$ 
    predlat ← predicted.lat ×  $\frac{\pi}{180}$ 
    predlon ← predicted.lon ×  $\frac{\pi}{180}$ 
  Calculate great circle distances
    dlon ← lon2 - lon1
    central_angle_actual ← ACOS((SIN(lat1) × SIN(lat2)) + (COS(lat1) × COS(lat2) ×
    COS(dlon)))
    central_angle_predicted ← ACOS((SIN(lat1) × SIN(predlat) + (COS(lat1) ×
    COS(predlat) × COS(predlon - lon1)))
  Calculate distances
    distance_actual ← R × central_angle_actual                                ▷ R is the Earth’s radius
    distance_predimated ← R × central_angle_predicted
  Calculate loss
    loss ← |distance_actual - distance_predicted|
  Return loss
end function

```

3.3. AIS Data Preprocessing

The data quality significantly influences the model results. Hence, this paper will present various preprocessing steps to improve the quality of the data. Currently, common trajectory data preprocessing operations include data cleaning, trajectory segmentation, trajectory interpolation, etc.

Data cleaning: Generally, AIS data can contain four types of errors or noise: abnormal MMSI column data, abnormal SOG column data, repeated data, and abnormal COG column data. The MMSI, the maritime mobile service identification code, is typically nine digits long. Abnormal MMSI column data indicate MMSI lengths that deviate from the standard length, as depicted in Figure 7a. The ship’s SOG represents the ship’s velocity. Rapid changes in SOG values are rated in practical cases and thus can be considered abnormal, as illustrated in Figure 7b. Repeated data in AIS may disturb the training of neural networks and lead to overfitting. Therefore, it is necessary to remove repeated data, as depicted in Figure 7c. COG indicates the ship’s direction. Its valid values range from 0 to 360. COG values outside this range are considered abnormal, as illustrated in Figure 7d.

Trajectory segmentation: When the ship encounters signal interference during AIS signal transmission, the received AIS data may exhibit irregular time intervals. To ensure more accurate and effective data analysis, segmentation of the data into sub-trajectories is essential. Segmentation points of the trajectory are determined based on the stop condition, observation interval, and speed condition. A ship is considered stopped when it remains within a 1 km radius for over 0.5 h and the distance between two points exceeds 5 km. The sub-trajectory is divided if there are no data for 0.5 h and the speed is below 1 knot (1.852 km/h).

Trajectory interpolation: Due to possible obstacles and electromagnetic interference, AIS data are often incomplete or incorrect. After removing certain abnormal data, trajectory repair is necessary to facilitate subsequent data analysis and mining. Mean interpolation and cubic spline interpolation are used for comparative repair. The results indicate that

cubic spline interpolation yields superior results; thus, we selected this method to repair trajectory data.

MMSI	BaseDateTime	LAT	LON	SOG	COG
316005715	2023-06-01T00:00:00	49.29717	−122.996	6.2	259.3
368158760	2023-06-01T00:00:00	29.67214	−91.1351	0	280.7
9110192	2023-06-01T00:00:01	21.3162	−157.879	0	136.7
367416420	2023-06-01T00:00:01	21.31005	−157.869	0.7	266.9

(a)

MMSI	BaseDateTime	LAT	LON	SOG	COG
319963000	2023-06-01T00:04:44	27.52827	−94.3162	11.6	359.2
319963000	2023-06-01T00:06:25	27.53372	−94.3162	11.6	359.6
319963000	2023-06-01T00:07:46	27.53809	−94.3163	11.6	359.1
319963000	2023-06-01T00:10:05	27.54554	−94.3159	11.5	363.0

(b)

MMSI	BaseDateTime	LAT	LON	SOG	COG
232046024	2023-06-01T00:10:55	25.89546	−80.128	1.1	360
232046024	2023-06-01T00:11:56	25.88725	−80.1358	32.3	219.4
232046024	2023-06-01T00:14:49	25.8952	−80.1284	2	230.6
232046024	2023-06-01T00:16:19	25.89557	−80.128	1.1	360

(c)

MMSI	BaseDateTime	LAT	LON	SOG	COG
215281000	2023-06-20T12:34:10	27.09598	−94.3187	3.1	56.2
215281000	2023-06-20T12:35:12	27.09645	−94.3179	3	59.2
215281000	2023-06-20T12:35:12	27.09645	−94.3179	3	59.2
215281000	2023-06-20T12:36:16	27.09694	−94.317	2.9	59.7

(d)

Figure 7. Four forms of erroneous and noisy impacts. (a) Abnormalities in the MMSI column; (b) irregularities in the COG column; (c) anomalies in the SOG column; (d) duplicate data.

4. Experimental Results and Analysis

In this study, we first introduce the dataset and then provide a brief explanation of the evaluation metrics and experimental settings. Finally, we evaluate the performance of each model and randomly select various types of trajectories to assess the model's generalization ability.

4.1. Description of Experimental Data

This study utilized AIS data that are publicly accessible via the Marinecadastre website [66], which provides free information on ship movements within US waters spanning from 2009 to 2023. Figure 8 illustrates a data visualization diagram of US coastal waters. For this study, AIS data from the pink box on the left of Figure 8 in June 2023 were selected, with a data size of 16.3 GB. After preprocessing (described in Section 3.2), the data were split into the training set, testing set, and validation set in a ratio of 8:1:1.

4.2. Experimental Evaluation Indicators

In this study, the mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE) are used to evaluate the validity of the model. Among them, MAE is used to measure the mean absolute difference between the predicted value and the actual value of the ship [67], while RMSE represents the square root of the mean squared difference between predicted and true values [68]. Smaller MAE and RMSE values indicate that the modeled trajectory has less deviation from the actual situation. MAPE measures the degree of deviation between the predicted values and the actual values [49]. A lower value of MAPE indicates a better fit between the predicted value curve and the true value curve, making it the most straightforward evaluation index.

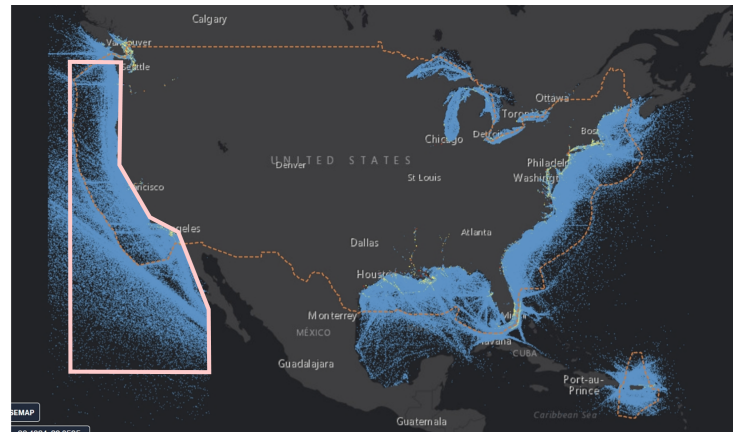


Figure 8. A map showing the area where the dataset was obtained.

4.3. Experimental Environment

All methods were implemented with PyTorch on Ubuntu 20.04 and trained using an NVIDIA RTX 2080 Ti GPU. During training, the same experimental parameters were used in all experiments. We used the CosineAnnealingLR learning rate to train the network, with an initial learning rate of 0.000005. The batch size was set to 16. To make better use of the GPU for training, we trained the framework with 150 epochs.

4.4. Comparison Between LSTM and Other Baseline Models

To establish an appropriate baseline model, the preprocessed AIS data were used as input for the LSTM model and compared with several classical time-series baseline models. The historical trajectory sequence of the preceding 18 min was utilized as input to predict the subsequent 2 min. The experimental results, detailed in Table 3, present the calculated RMSE, MAE, and MAPE values for LON and LAT. The conclusions drawn from these experiments are as follows:

GRU [63]: another type of RNN. The overall index accuracy of the GRU model is low in terms of RMSE, MAE, and MAPE, which are an order of magnitude different from other networks' errors. This is because the model performs poorly with the data in this particular case. The GRU model struggles to adequately extract key features of a ship's trajectory when processing spatiotemporal information. This difficulty may be due to the model's suboptimal performance in this particular scenario. The relatively high MAPE value of the GRU indicates that the percentage deviation of its predicted results from the actual values is rather large, thereby demonstrating an evident deficiency in accuracy.

RNN [23]: recurrent neural network. The experimental results presented in this article indicate that, in comparison to the LSTM model, simpler networks tend to yield more accurate results. Hence, this article compares the earliest RNN model. However, from the observed phenomenon in Table 3, the RNN exhibits significant differences across various performance metrics, particularly in terms of RMSE and MAE. This discrepancy may be attributed to the absence of a gating mechanism, such as that found in LSTM, which effectively manages long-term dependencies within extensive time-series data. Consequently, this limitation can lead to an increase in prediction errors. The MAPE value of the RNN is higher than that of LSTM, which indicates that, in terms of relative error, the prediction effect of the RNN is not as good as that of LSTM, and its predicted values deviate from the actual values to a greater extent.

TCN [58]: a network architecture specifically designed for processing time-series data that deals with sequence dependencies by using convolutional layers instead of cyclic layers. The TCN network is expected to exhibit robust time memory capabilities. However, despite its strong time memory ability, the TCN network does not perform as well as LSTM across various indicators, with overall accuracy being notably lower. The suboptimal performance of the TCN network stems from its inability to fully extract the features present

in the data. The MAPE value of TCN in longitude prediction is much higher than that of LSTM, and it is also not as good as LSTM in latitude prediction. This shows that TCN is not as good as LSTM in controlling relative errors, which limits its prediction accuracy.

Seq2seq [69]: Overall, in longitude prediction, it is close to some models with better performance. However, in latitude prediction, there is a significant gap compared with LSTM. This indicates that the Seq2seq model has certain limitations when processing latitude information in these data. It is likely that some crucial information has been lost during the encoding–decoding process, leading to a decline in prediction accuracy. The relatively high MAPE value of Seq2seq in latitude prediction shows that when it comes to latitude prediction, the percentage deviation of its predicted values from the actual values is rather large, and the relative error is quite evident.

BiLSTM [70]: a bidirectional network. The prediction accuracy of the BiLSTM model closely matches that of the LSTM model, yet it remains slightly lower. The introduction of the bidirectional structure may add unnecessary complexity. Although BiLSTM can view past and future information simultaneously, this also causes information leakage, causing the model to perform poorly in actual prediction. The MAPE value of BiLSTM is higher than that of LSTM, which means that BiLSTM is not as good as LSTM in controlling the relative error, and its predicted results deviate to a greater degree from the actual values.

Table 3. Comparison results of models. ↓ means that lower is better. Note: The content shown in bold in the table represents the model used in this paper and its corresponding relevant indicator data.

Model	LAT_MAE↓	LON_MAE↓	LON_RMSE↓	LAT_RMSE↓	LON_MAPE	LAT_MAPE
GRU	0.92702	0.39629	0.46053	1.02917	0.00151	0.02301
RNN	0.26230	0.04423	0.07378	0.38184	0.00152	0.02241
TCN	0.17887	0.02856	0.04268	0.27326	0.00488	0.02486
Seq2seq	0.16156	0.02574	0.05545	0.35212	0.00173	0.00794
BiLSTM	0.19081	0.01726	0.03231	0.44305	0.00803	0.00423
LSTM	0.10134	0.01447	0.03203	0.15862	0.001424	0.001903

4.5. Ablation Study

The experimental results in the previous section confirmed the effectiveness of the LSTM model for the data analyzed in this paper. Subsequently, we proceeded to enhance this model further based on its performance, aiming to tailor it more precisely to the task at hand. To dissect the contribution of each module following the enhancement, we conducted an ablation experiment.

To scrutinize the contribution of each module following the improvement, this section analyzes the results of the ablation experiment. To this end, different modules were removed, and the TPALSTM Network, LSTM TPCA (+) network, and TPALSTM were constructed. Additionally, we introduced the TPCA (+) network and our model (GCRLoss (+)). Figure 9 illustrates the error/accuracy of the five models on the validation set for each indicator. As shown in Figure 9, the error decreases gradually with increasing iterations. Similarly, removing any module from the model leads to a decline in its accuracy. Moreover, the addition of the GCRLoss function significantly reduces the error, suggesting that the Great Circle Route Loss Function is more suitable for this application scenario and markedly enhances the model's performance.

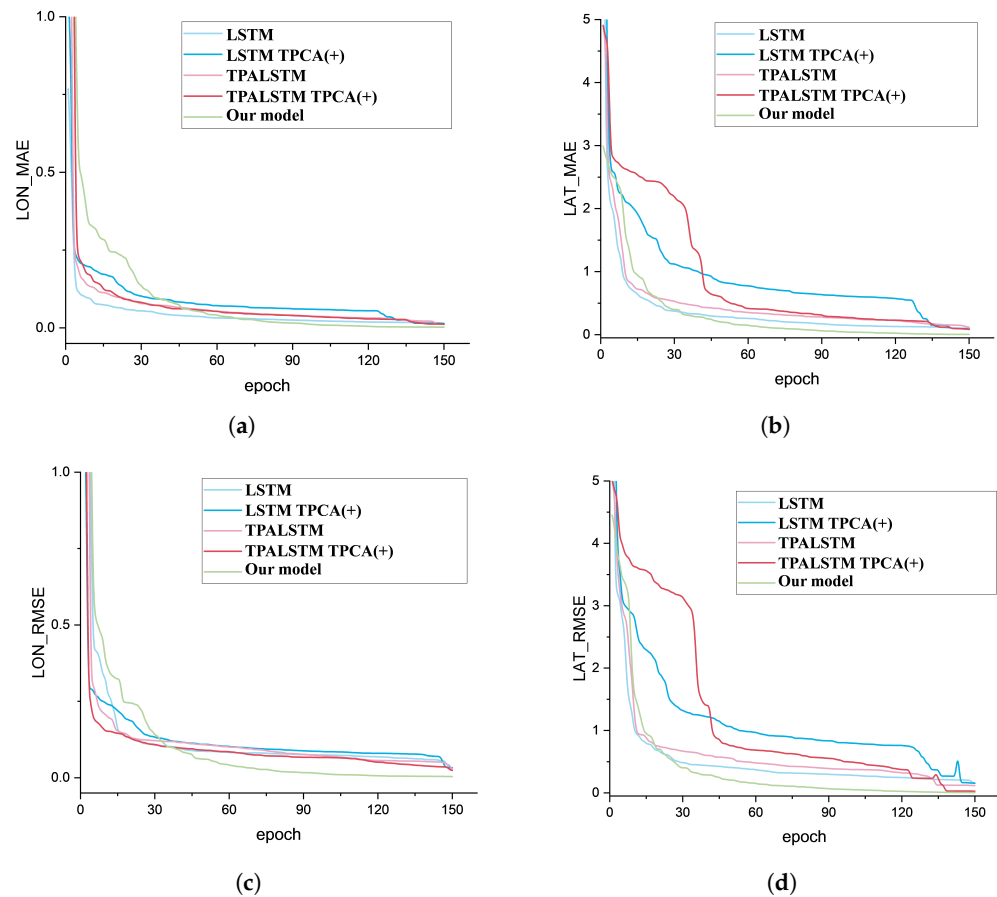


Figure 9. A comparison of the model results based on various indicators during the training process. (a) MAE of LAT. (b) MAE of LON. (c) RMSE of LAT. (d) RMSE of LON.

Table 4 displays the experimental results on the test set, showcasing the MAE, MAPE, and RMSE values of longitude and latitude measured in the subsequent 2 min. As evident from the table, the model's accuracy improves progressively as the number of modules increases. Remarkably, excluding the GCRLoss function leads to a significant performance decrease in the model. These findings from the ablation experiment confirm the indispensability of each module in the model proposed in this paper, demonstrating that each module contributes significantly to enhancing the overall model accuracy.

Table 4. Prediction results of ablation experiments on the test set. ↓ means that lower is better. Note: ✓ indicates that the corresponding module participates in the experiment under the experimental setting of that row. The content shown in bold in the table represents the model used in this paper and its corresponding relevant indicator data.

LSTM	TPA	TPCA	GCRLoss	LAT_MAE↓	LON_MAE↓	LON_RMSE↓	LAT_RMSE↓	LON_MAPE	LAT_MAPE
✓				0.05134	0.03447	0.04265	0.07862	0.00142	0.00190
✓	✓			0.04805	0.03394	0.03943	0.06105	0.00126	0.00135
✓		✓		0.02175	0.01409	0.03252	0.05435	0.00114	0.00126
✓	✓	✓		0.01292	0.01191	0.02404	0.02749	0.00105	0.00114
✓	✓	✓	✓	0.001084	0.002023	0.002705	0.001687	0.00093	0.00057

4.6. Comparison Study

This paper aims to assess our model's accuracy through comparative experiments with several models, namely, MTAD_GAT, LSTNet, TPALSTM, and STGCN.

MTAD_GAT [71]: This approach utilizes a Graph Attention Network (GAT) to capture both feature and temporal dependencies. These are integrated into a streamlined GRU network, enhancing detection efficiency while minimizing computational overhead.

LSTNet [72]: This multivariate time-series prediction method extracts features from time-series data through the convolution layer to capture short-term local information, and LSTM is used to capture long-term macro information.

TPALSTM [65]: This approach introduces a novel attention mechanism for identifying pertinent time series and utilizing their frequency-domain information in multivariate prediction.

STGCN [73]: This Graph Convolutional Neural Network (GCN) excels in handling spatiotemporal data and effectively captures dependencies across spatial and temporal dimensions in ship trajectory data. Implementing STGCN for ship trajectory prediction enables accurate forecasting of future paths.

4.6.1. Single-Step Trajectory Prediction

We assess the effectiveness of various algorithms for single-step prediction. The experimental outcomes are detailed in the initial row of Table 5, highlighting the optimal results in bold. Table 5 indicates that our method outperforms the other algorithms in MAE, MAPE, and RMSE. On the whole, compared with the other four methods, our method achieves superior performance in all attributes, indicating that incorporating geographic knowledge into the model can improve accuracy.

Specifically, our method achieves much better performance than MTAD_GAT and STGCN. Our method's MAE decreased by approximately 0.07 and 0.08 in LON and LAT predictions, respectively. Concurrently, its RMSE increased by approximately 0.02 and 0.09 in LON and LAT predictions, respectively. Moreover, the accuracy of SOG and COG also improved significantly. While MTAD_GAT and STGCN effectively learn stable local attribute correlations, they overlook global time-varying features and inter-attribute correlations. Our method addresses these shortcomings by capturing both time patterns and global inter-attribute relationships, as demonstrated by the experimental results.

Our method also outperforms TPALSTM and LSTNet. Our method reduces MAE and RMSE by about 0.03 and 0.035, respectively, in LON prediction. Our method improves MAE and RMSE by 0.04 and 0.05, respectively, in LAT prediction. In SOG and COG prediction, great improvements are also made with our method. The other methods incorporate information from all attributes when extracting features from AIS data, but they neglect to consider the relationships between attributes, local feature variations, and global time-dependent correlations. As a result, they fail to capture a rich and accurate representation of the trajectory, leading to suboptimal prediction accuracy. Based on the MAPE results, the predictive model demonstrated superior performance compared to the other models across all four variables.

4.6.2. Multi-Step Trajectory Prediction

In this section, the robustness and stability of the proposed method are evaluated by analyzing multi-step predictions of ship trajectory points with varying prediction step sizes. Specifically, predictions are conducted for 5, 10, 15, and 20 steps, which approximately correspond to durations of 10, 20, 30, and 40 min, respectively. Additionally, comparisons with other trajectory prediction methods are made. Table 5 indicates that for short-term and medium-term ship trajectory predictions, the model proposed in this paper achieves the best performance in terms of all metrics. It is found that as the prediction step size increases, the accuracy of the model declines, and greater differences tend to emerge among different prediction methods. When the prediction step is 5, each model can accurately predict the future trajectory in the initial few steps. Overall, the prediction accuracy of

longitude is generally higher than that of latitude. This is probably because latitude data vary significantly, and ships mainly head in the east–west direction.

Table 5. Comparison results of models. Note: The content shown in bold in the table represents the model used in this paper and its corresponding relevant indicator data.

Prediction Step	Methods	MAE				RMSE				MAPE			
		LON	LAT	SOG	COG	LON	LAT	SOG	COG	LON	LAT	SOG	COG
1	MTAD_GAT	0.086336	0.090204	0.852586	2.962924	0.101684	0.11592	1.262215	12.924484	0.007599	0.007341	0.291612	0.49913
	TPALSTM	0.033944	0.048052	0.759806	4.833171	0.039438	0.061051	1.203548	5.459395	0.005436	0.007561	0.46512	0.06241
	STGCN	0.076652	0.037563	0.388463	1.338905	0.049328	0.110829	0.901796	4.245331	0.004365	0.008452	0.36541	0.23415
	LSTNet	0.04332	0.027212	0.792366	2.314438	0.054102	0.035895	1.231251	4.789644	0.003637	0.00924	0.3008	0.02218
	Our model	0.002023	0.004536	0.069415	0.191063	0.002705	0.001687	0.097449	1.277275	0.00093	0.00057	0.15714	0.01534
5	MTAD_GAT	0.222017	0.16672	1.135313	9.318908	0.226321	0.189552	1.523652	13.14574	0.010712	0.00554	0.2073	0.0895
	TPALSTM	0.071926	0.061051	1.121298	21.207679	0.077599	0.076011	1.539583	21.93227	0.08914	0.02315	0.45128	0.07251
	STGCN	0.0085682	0.045327	0.756145	2.452145	0.062345	0.135675	1.231468	6.45247	0.061424	0.014557	0.321454	0.561447
	LSTNet	0.055742	0.035895	1.251253	3.147069	0.078591	0.052041	1.699103	7.046304	0.00465	0.01085	0.59653	0.02509
	Our model	0.004437	0.006278	0.084564	0.245123	0.005393	0.006556	0.101414	2.34561	0.001611	0.00715	0.56562	0.21331
10	MTAD_GAT	0.356247	0.292136	1.384505	15.296236	0.361537	0.328295	1.764433	16.421535	0.01172	0.004844	0.60732	0.08762
	TPALSTM	0.105364	0.070332	1.357819	38.045608	0.111582	0.086357	1.742947	38.767856	0.006156	0.034512	0.15482	0.54561
	STGCN	0.100562	0.067568	0.945124	3.122152	0.110435	0.153682	1.452651	7.154201	0.024615	0.021467	0.54154	0.782156
	LSTNet	0.079526	0.05133	1.590686	4.524153	0.109809	0.077114	1.988238	8.264787	0.006869	0.01619	0.67532	0.03754
	Our model	0.005516	0.01984	0.104214	0.414135	0.007103	0.019978	0.241574	3.541244	0.001912	0.008912	0.663923	0.324652
15	MTAD_GAT	0.467683	0.401736	1.504346	20.636161	0.474728	0.456042	1.9107316	21.729131	0.01206	0.007062	0.81273	0.09647
	TPALSTM	0.128553	0.082603	1.424813	51.818588	0.135486	0.10622	1.836279	52.725833	0.0094514	0.045713	0.71545	0.76411
	STGCN	0.113561	0.083214	1.123541	5.215431	0.130546	0.175313	1.624154	9.145414	0.021342	0.025417	0.45713	0.45771
	LSTNet	0.093733	0.072858	1.695865	6.660522	0.129841	0.105266	2.116277	12.752275	0.007937	0.01228	0.74391	0.050962
	Our model	0.007983	0.036589	0.124561	0.647854	0.008991	0.036784	0.475864	4.751432	0.002501	0.018947	0.791012	0.42312
20	MTAD_GAT	0.560471	0.471043	1.677445	25.434125	0.568225	0.541822	2.104631	26.647263	0.012513	0.009571	0.68632	0.58895
	TPALSTM	0.143661	0.105757	1.569451	63.011222	0.15162	0.141545	1.982249	64.211448	0.023548	0.052146	0.784153	0.85461
	STGCN	0.123562	0.100213	1.144241	6.854241	0.157521	0.183564	1.741251	10.452114	0.012358	0.036784	0.84513	0.86571
	LSTNet	0.100393	0.098265	1.739334	8.69294	0.135451	0.134256	2.205585	14.064968	0.008370	0.03106	0.79853	0.05139
	Our model	0.014342	0.054848	0.345814	0.847475	0.014852	0.055091	0.674251	5.954145	0.00451	0.002931	0.81563	0.49957

Box plots illustrate the MAE and RMSE for the latitude and longitude attributes with a step length of 20, allowing for a comparison of the stability of the four models, as shown in Figure 10. In this scenario, smaller box plot lengths indicate higher model stability. It is observed that, except for MTAD_GAT, the models exhibit relatively stable performance, with our method demonstrating the highest stability.

In summary, opting for LSTM as the foundational model for this method offers clear advantages over other basic models, notably by significantly enhancing the accuracy.

4.6.3. Comparison of Different Types of Trajectory Predictions

To verify the model's generalization ability, various types of trajectories were selected for prediction. When a ship is sailing in a straight line, its actual trajectory is along a route that is very similar to a straight line. The target ship begins to sail according to the planned route, while the sailing direction and speed remain in a relatively stable state. As depicted in Figure 11, the predicted trajectory also appears as a straight line, exhibiting a high degree of similarity to the true value.

When the route exhibits curvature, the trajectory prediction accuracy tends to be higher when the ship's SOG changes slowly, as depicted in Figure 12a. In such cases, the position prediction is closer to the real data, and the accuracy is higher. However, when SOG changes abruptly, as shown in Figure 12b, although the overall error of the model prediction increases, the overall trend of the predicted route remains largely consistent with the real route. As shown in Figure 12b, when a ship suddenly encounters bad weather, such as strong winds, heavy rain, or rough seas, or receives a new turn order and thus drastically changes its course, its trajectory will show a step change. Although the overall

prediction results of our model are less biased, this situation exposes a significant limitation of the current approach: it does not take into account unexpected factors, such as weather and sea state. Future research should focus on integrating real-time weather data, sea state information, and potential changes in navigation orders into the prediction model.

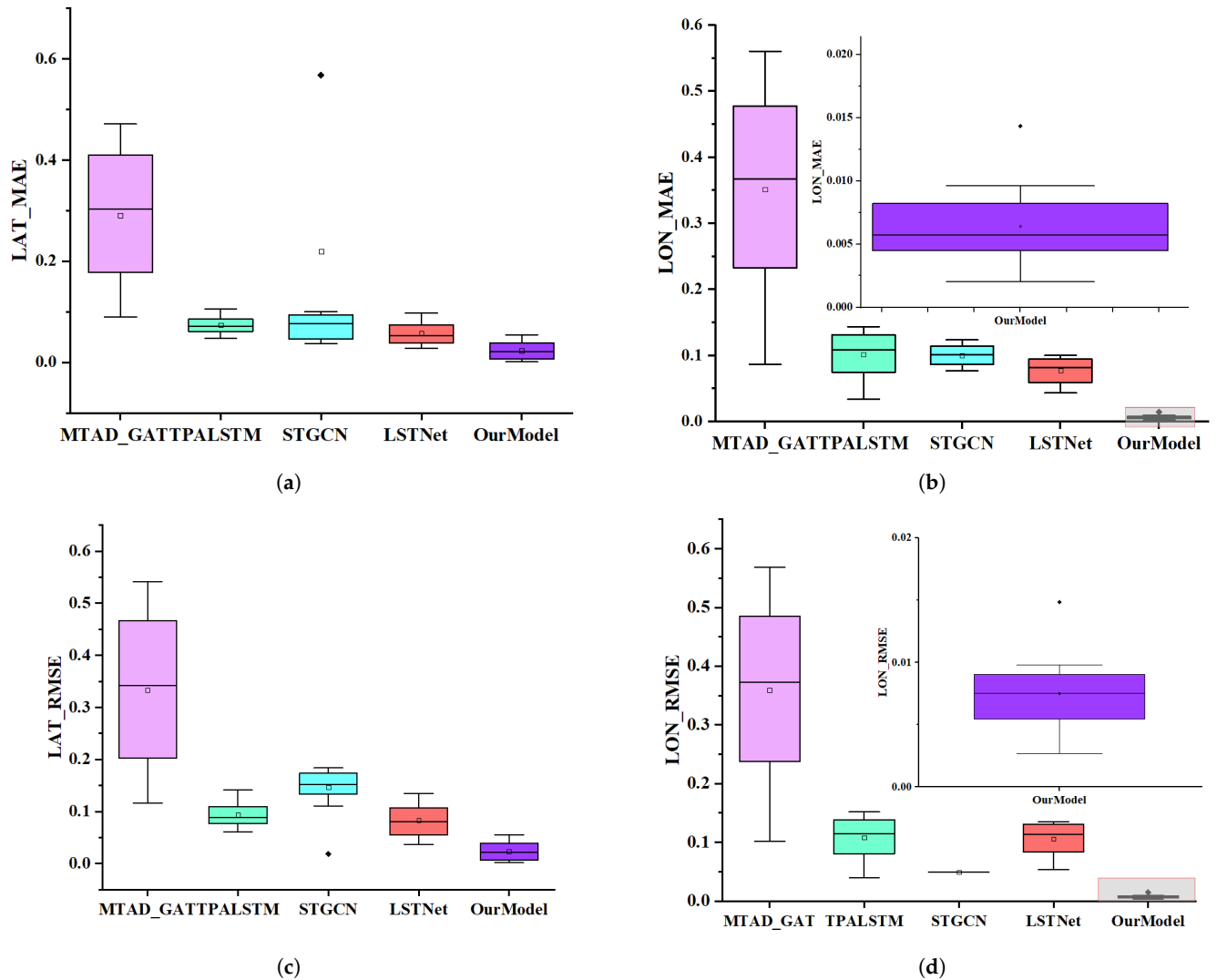


Figure 10. Box plots of prediction MAE and RMSE with a length of 20 steps. (a) LAT prediction MAE. (b) LON prediction MAE. (c) LAT prediction RMSE. (d) LON prediction RMSE. Note: The circles inside the bars represent abnormal values, and the squares represent the mean values.

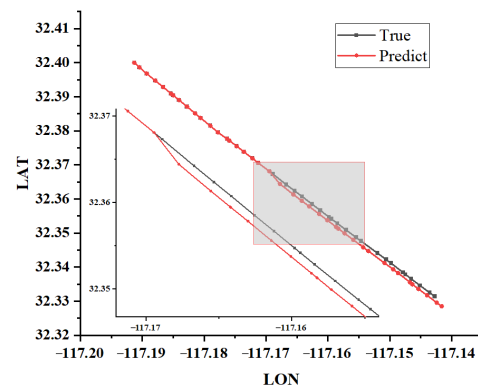


Figure 11. Prediction results of the proposed model when sailing in a straight line.

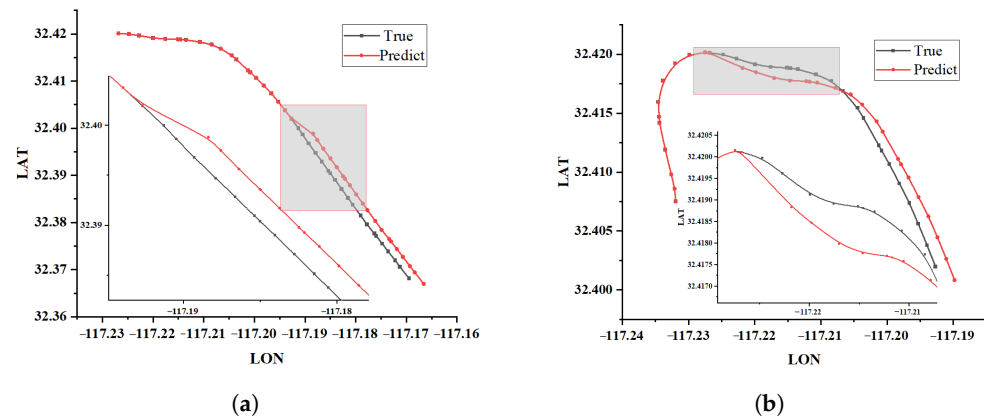


Figure 12. Prediction results of the proposed model for curved navigation with different SOG changes. (a) Prediction results with slow SOG changes. (b) Prediction results with rapid SOG changes.

An experiment comparing different prediction methods was conducted under curved navigation conditions, with the results depicted in Figure 13. The experimental findings reveal that both the proposed model and the MTAD_GAT model exhibit prediction trends consistent with the direction of the true trajectory. However, the proposed model demonstrates superior prediction accuracy compared to the MTAD_GAT model. Although the LSTNet model's prediction trend may not be as robust as the others, it ranks second in prediction error accuracy behind the proposed model. Notably, since TPALSTM represents an improvement and optimization based on the LSTM model, this outcome underscores the significant advancements achieved by the proposed model. In contrast, the error propagation of the STGCN model is smaller and the same as the LSTNet forecast trend. It is worth noting that due to error propagation in the model prediction process, the trajectory prediction result often exceeds the true trajectory length.

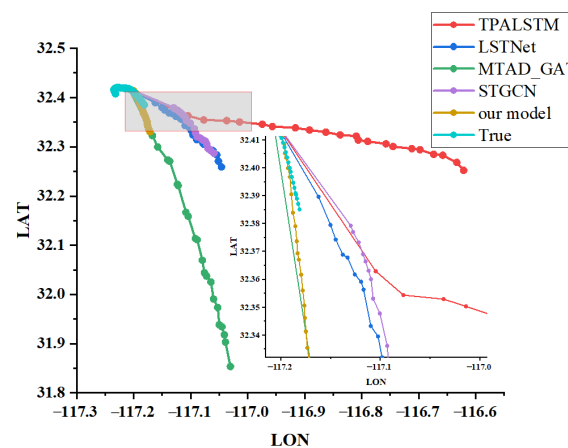


Figure 13. Comparison results of curved navigation trajectory prediction.

This study also examined the predictions of longitude and latitude across various time steps, as depicted in Figure 14. The analysis of the experimental outcomes indicates that our newly proposed model typically surpasses other models in prediction accuracy. However, it is noted that the improvement in predicting latitude is less significant compared to the enhancement in longitude predictions. This discrepancy can be attributed to the significant variations in trajectory observed in the latitude dimension, as previously analyzed in the discussion.

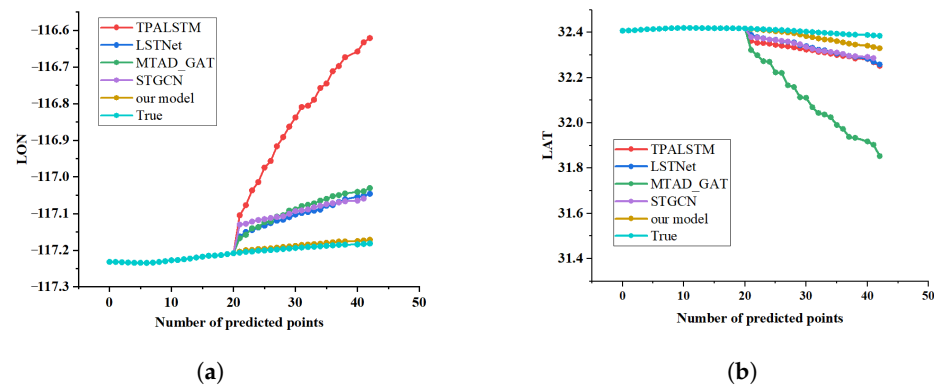


Figure 14. Comparison of curved navigation prediction results. (a) Longitude prediction results; (b) latitude prediction results.

5. Conclusions

5.1. Discussion

Ship trajectory prediction is of great significance for strengthening maritime management, reducing the incidence of maritime traffic accidents, and assisting maritime departments in making appropriate decisions. With the widespread popularization of AIS devices, vast amounts of AIS data have been generated, from which the latent information on ship motion can be effectively extracted and applied in the field of ship trajectory prediction. Deep learning algorithms have demonstrated certain advantages in the field of ship trajectory prediction. However, for several key features involved in trajectory data, such as longitude, latitude, speed, and course, there are still some pressing problems to be solved. On the one hand, deep learning algorithms still have deficiencies in handling the long-term dependencies and dynamic correlations among these features. On the other hand, fully integrating potential geographic information into the neural network also poses a significant challenge. Given this, this paper proposes a series of targeted solutions. Specifically, the TPCA mechanism is elaborately designed to deeply explore the local trend characteristics in the process of trajectory point changes. The TPA mechanism is introduced to accurately capture the dynamic feature correlations among multiple variables across different time steps. The GCRLoss is constructed to deepen the understanding of spatial relationships and geographic information. Through the coordinated operation of these mechanisms, efforts are made to comprehensively improve the accuracy and comprehensiveness of ship trajectory prediction.

In terms of model performance evaluation, this paper uses the RMSE, MAPE, and MAE as evaluation metrics to conduct a comprehensive and objective comparison of the performance of different models. Using relevant data for the US West Coast, first, a rigorous screening and preprocessing procedure was carried out. Then, various classic deep learning-based time-series prediction models were included in the comparison study to screen out a benchmark model with better compatibility with the data in this paper. After a series of verification experiments, the results showed that the LSTM model's most outstanding performance was obtained when dealing with the data involved in this paper, and its prediction error was the lowest among all the compared models. Based on this, the method proposed in this paper uses the LSTM model as its foundation. It underwent targeted improvements and optimizations in terms of feature extraction and attention mechanisms according to the characteristics of AIS data. From the results of the ablation experiment, each module shows relatively high performance when dealing with such data. Further, through comparative experiments with various other prediction models, it is found that, whether in the context of single-step prediction or multi-step prediction, the prediction results generated by the method proposed in this paper have a higher degree of conformity with the actual trajectories of ships. Specifically, the experimental data show that when predicting small-angle and large-angle trajectories, the prediction results of the model proposed in this paper present relatively small error values in terms of three evaluation

metrics, RMSE, MAE, and MAPE, fully demonstrating the superiority and effectiveness of the method in ship trajectory prediction. Accurate trajectory prediction methods can contribute to practical applications such as ship navigation assistance, maritime traffic management, port scheduling optimization, and maritime search and rescue support.

Of course, the method proposed in this paper also has certain limitations. One of the issues is its relatively high computational complexity. When implementing the relevant components, such as TPCA and TPA, numerous complex computational processes and operational steps are involved. Secondly, although this method attempts to integrate potential geographic information through GCRLoss to better consider the impact of geographic factors on ship trajectories, in actuality, the data on which this paper is based are mainly track data generated by ships under normal navigation conditions. It does not fully take into account various geographic features (different channel terrains, shoal distributions, etc.), changeable ocean currents, and complex weather conditions (storm surges, sea ice, etc.).

5.2. Future Work

To further enhance the performance and applicability of our model, several directions for future work have been identified. Firstly, extensive testing of the model in various real maritime situations is necessary. This should cover a wide range of combinations of different external factors, including weather changes, interactions between ships, and geographical differences. By subjecting the model to these diverse scenarios, we can gain a better understanding of its performance limitations and make the necessary adjustments to improve its robustness and accuracy in real-world applications. Secondly, we need to make full use of the channel information in historical data. Specifically, we will set up test scenarios under different channel conditions, such as varying water flow speeds and degrees of channel curvature. This will involve developing appropriate methods to extract and incorporate the relevant channel information into the prediction model, thereby improving the comprehensiveness of our predictions. Finally, we aim to explore the use of additional data sources for ship trajectory prediction. In particular, we will look into the integration of shore-based radar data, as well as the combination of trajectory prediction with SAR data, visible light data, and other multi-source data for collaborative analysis. This multi-data approach will provide more comprehensive geographic information about ship targets, enhancing our situational awareness of key marine targets and ultimately leading to more accurate and reliable ship trajectory predictions.

Author Contributions: Conceptualization, Y.Z., H.G., J.L., Z.G., D.Y. and L.D.; methodology, Y.Z.; validation, Y.Z. and H.G.; formal analysis, Y.Z. and H.G.; investigation, Y.Z. and H.G.; resources, Y.Z. and H.G.; data curation, Y.Z. and H.G.; writing—original draft preparation, Y.Z., H.G., J.L., Z.G., D.Y. and L.D.; writing—review and editing, Y.Z. and H.G. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the National Natural Science Foundation of China under Grant 42201443.

Data Availability Statement: Data were obtained from <https://marinecadastre.gov/ais/> (accessed on 15 July 2023).

Conflicts of Interest: The authors declare no conflicts of interest.

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